COMPREHENSIVE BUSINESS ANALYTICS REPORT

Superstore Sales Dataset — train.csv | Prepared by Amiin Mohamed

Prepared by:	Amiin Mohamed
Role:	Lead Data Analyst
Dataset:	train.csv — Superstore Sales (Cleaned)
Records:	9,800 rows × 18 columns
Period:	January 2015 – December 2018
Report Date:	June 2026

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SECTION 1 — KEY PERFORMANCE INDICATORS

TOTAL SALES	TOTAL ORDERS
$2,261,536.55	**4,922**
Sum of all revenue	Distinct Order IDs

AVG ORDER VALUE	UNIQUE CUSTOMERS
$459.48	**793**
Total Sales / Total Orders	Distinct Customer IDs

TOP REGION	TOP SEGMENT
West	**Consumer**
$710,219.60 in sales	$1,148,060.29 in sales

TOP CATEGORY	TOP PRODUCT
Technology	**Canon 2200 Copier**
$827,455.86 in sales	$61,599.83 in sales

KPI DETAIL TABLE

#	KPI	Value	Note
1	**Total Sales**	$2,261,536.55	Driven by Technology & Furniture
2	**Total Orders**	4,922	High volume across 4 years
3	**Avg Order Value**	$459.48	Improvable through bundling
4	**Unique Customers**	793	Loyalty programs recommended
5	**Top Region**	West — $710,219.60	31.4% of total revenue
6	**Top Segment**	Consumer — $1,148,060.29	50.8% of all revenue
7	**Top Category**	Technology — $827,455.86	36.6% — highest density
8	**Top Product**	Canon imageCLASS 2200 Copier	$61,599.83 — star product

SECTION 2 — EDA: SHIP MODE & CUSTOMER SEGMENT

Analysis of sales distribution across shipping methods and customer segments to optimize logistics and target high-value groups.

Sales by Ship Mode

Ship Mode	Total Sales ($)	Orders	Avg Sale ($)	% Revenue
Standard Class	$1,340,831.40	5,859	$228.85	59.3%
Second Class	$449,914.00	1,902	$236.55	19.9%
First Class	$345,572.15	1,501	$230.23	15.3%
Same Day	$125,219.00	538	$232.75	5.5%
TOTAL	**$2,261,536.55**	**9,800**	**$229.93**	**100%**

Key Finding: Standard Class dominates (59.3%). Same Day shipping is severely underutilized at only 5.5% — premium clients likely want faster delivery.

Sales by Customer Segment

Segment	Total Sales ($)	Orders	Avg Sale ($)	% Revenue
Consumer	$1,148,060.29	5,101	$225.07	50.8%
Corporate	$688,494.02	2,953	$233.15	30.4%
Home Office	$424,982.24	1,746	$243.40	18.8%

Key Finding: Consumer = 50.8% of revenue but lowest AOV ($225.07). Home Office has highest AOV ($243.40) — premium clients worth targeting for upsell.

SECTION 3 — EDA: MONTHLY SALES TREND (2015–2018)

Annual Sales Summary

Year	Total Sales ($)	YoY Growth	Status
2015	$479,856.18	—	Baseline
2016	$459,435.87	-4.3%	Declining
2017	$600,192.68	+30.6%	Growing
2018	$722,051.82	+20.3%	Growing

2018 Monthly Breakdown

Month	Sales ($)	vs Peak	Performance
2018-01	$43,476.47	37%	Moderate
2018-02	$19,920.98	17%	Moderate
2018-03	$58,863.41	50%	Moderate
2018-04	$35,541.89	30%	Moderate
2018-05	$43,825.95	37%	Moderate
2018-06	$48,190.70	41%	Moderate
2018-07	$44,825.11	38%	Moderate
2018-08	$62,837.81	53%	Strong
2018-09	$86,152.88	73%	Strong
2018-10	$77,448.12	66%	Strong
2018-11	$117,938.13	100%	PEAK MONTH
2018-12	$83,030.37	70%	Strong

Key Finding: November 2018 was the peak month. Q4 consistently outperforms. See the Monthly Trend chart in Section 6.

SECTION 4 — EDA: GEOGRAPHIC ANALYSIS

Sales by Region

Region	Total Sales ($)	% of Total
West	$710,219.60	31.4%
East	$669,518.79	29.6%
Central	$492,646.78	21.8%
South	$389,151.38	17.2%

Top 10 States by Revenue

Rank	State	Total Sales ($)	% of National Revenue
1	**California**	$446,306.43	19.7%
2	**New York**	$306,361.07	13.5%
3	**Texas**	$168,572.40	7.5%
4	**Washington**	$135,206.87	6.0%
5	**Pennsylvania**	$116,276.72	5.1%
6	**Florida**	$88,436.50	3.9%
7	**Illinois**	$79,236.52	3.5%
8	**Michigan**	$76,136.07	3.4%
9	**Ohio**	$75,130.41	3.3%
10	**Virginia**	$70,636.72	3.1%

Key Finding: California + New York = 33.2% of all revenue. South (17.2%) is the most underserved region. See chart in Section 6.

SECTION 5 — EDA: SUB-CATEGORY PERFORMANCE

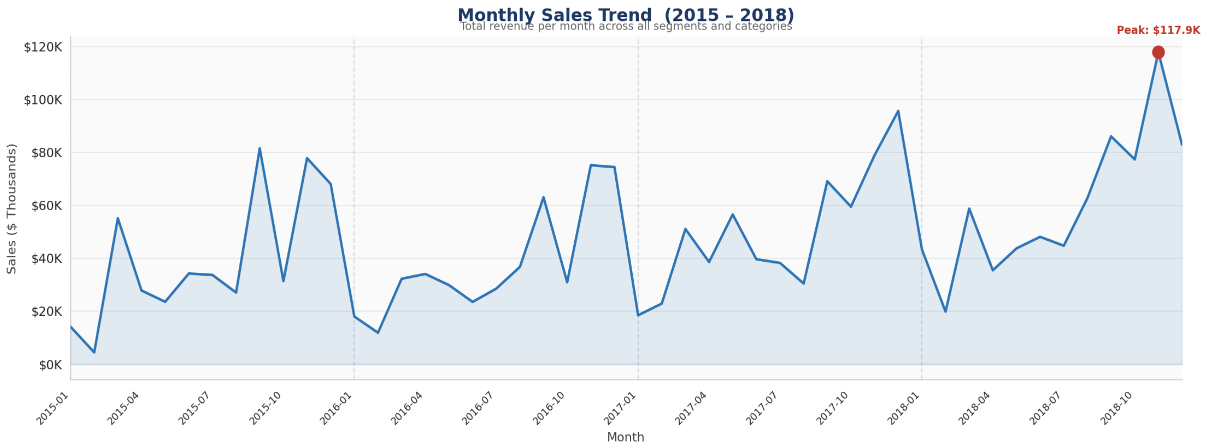
#	Sub-Category	Sales ($)	% Share	Performance
1	**Phones**	$327,782.49	14.5%	Top Performer
2	**Chairs**	$322,822.72	14.3%	Top Performer
3	**Storage**	$219,343.37	9.7%	Top Performer
4	**Tables**	$202,810.62	9.0%	Strong
5	**Binders**	$200,028.73	8.8%	Strong
6	**Machines**	$189,238.60	8.4%	Strong
7	**Accessories**	$164,186.70	7.3%	Good
8	**Copiers**	$146,248.07	6.5%	Good
9	**Bookcases**	$113,813.18	5.0%	Average
10	**Appliances**	$104,618.38	4.6%	Average
11	**Furnishings**	$89,211.98	3.9%	Average
12	**Paper**	$76,828.34	3.4%	Average
13	**Supplies**	$46,420.29	2.1%	Below Avg
14	**Art**	$26,705.42	1.2%	Low
15	**Envelopes**	$16,128.02	0.7%	Low
16	**Labels**	$12,347.71	0.5%	Needs Attention
17	**Fasteners**	$3,001.93	0.1%	Needs Attention

Best: Phones + Chairs = 28.8% of revenue. **Worst:** Fasteners + Labels + Envelopes = only 1.4% combined. Full chart in Section 6.

SECTION 6 — DATA VISUALIZATIONS

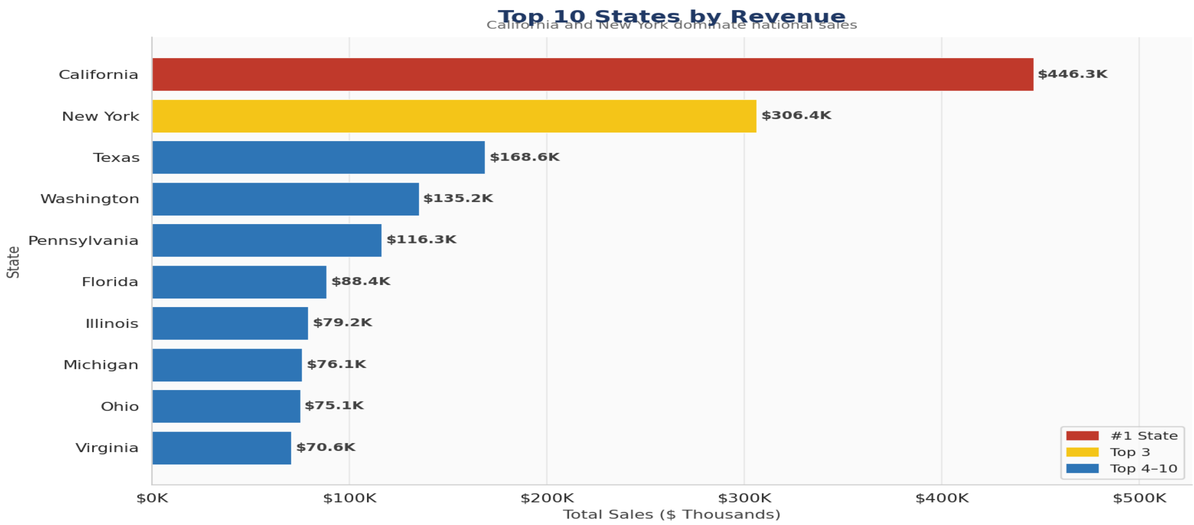
All charts below were generated using Python (Matplotlib & Seaborn) directly from the cleaned train.csv dataset. Each chart is titled, axis-labeled, and color-coded for clarity.

Chart 1 — Monthly Sales Trend (2015–2018)



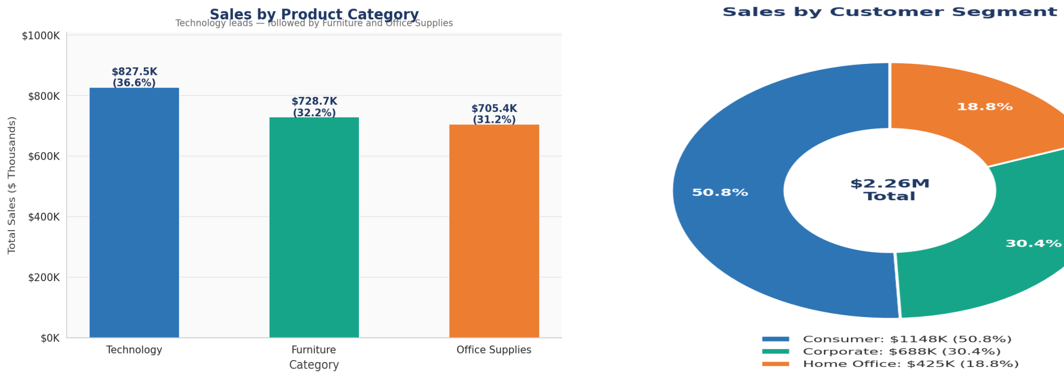
Line chart showing monthly revenue across 48 months. Red dot marks the peak month (November 2018 — $117.9K). Dashed lines mark year boundaries.

Chart 2 — Top 10 States by Revenue



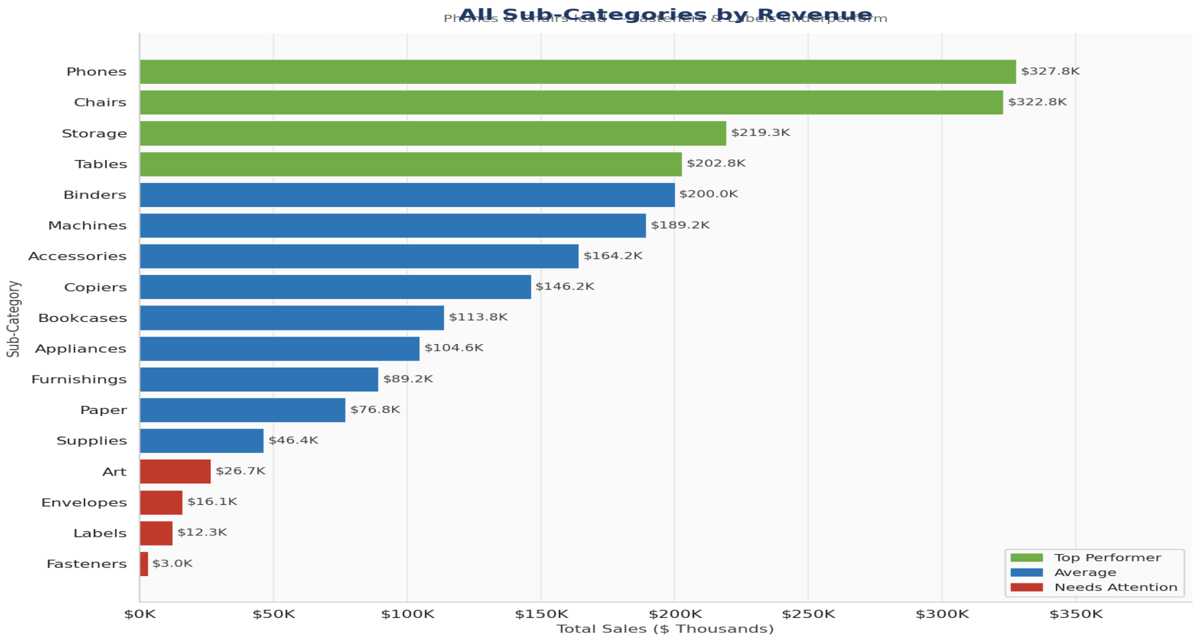
Horizontal bar chart — California leads with $446K (19.7% of national revenue). Red = #1, Gold = Top 3, Blue = Top 4–10.

Chart 3 — Sales by Category | Chart 4 — Sales by Customer Segment



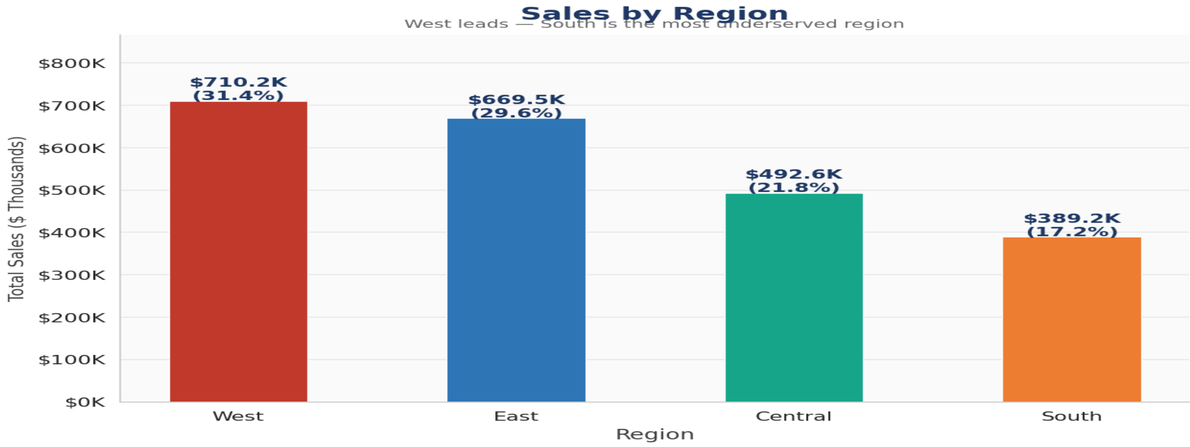
Left: Technology leads all categories at $827K (36.6%). Right: Consumer segment holds 50.8% of total revenue ($1.15M).

Chart 5 — All Sub-Categories by Revenue (17 sub-categories)



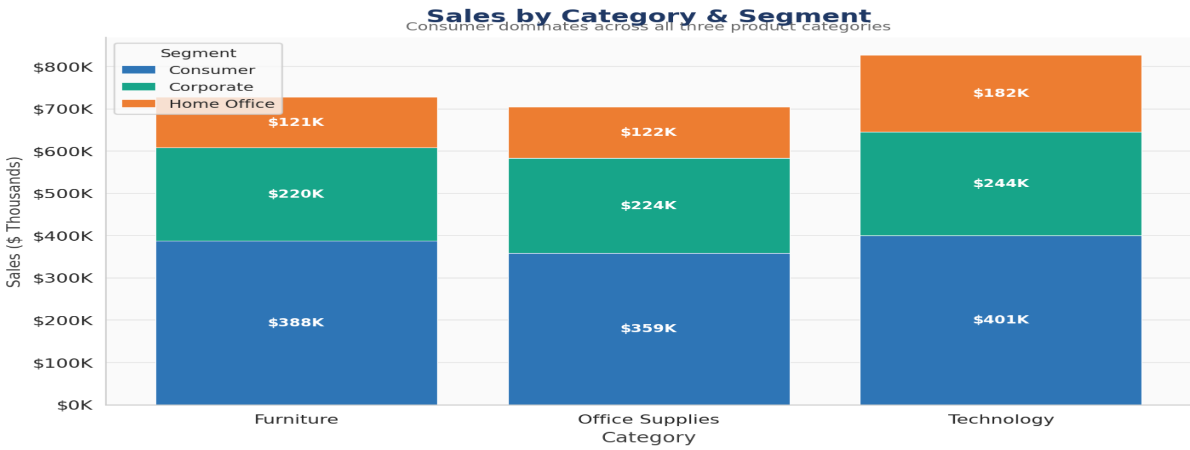
Green = Top Performer, Blue = Average, Red = Needs Attention. Phones and Chairs lead. Fasteners is the weakest at only $3K.

Chart 6 — Sales by Region



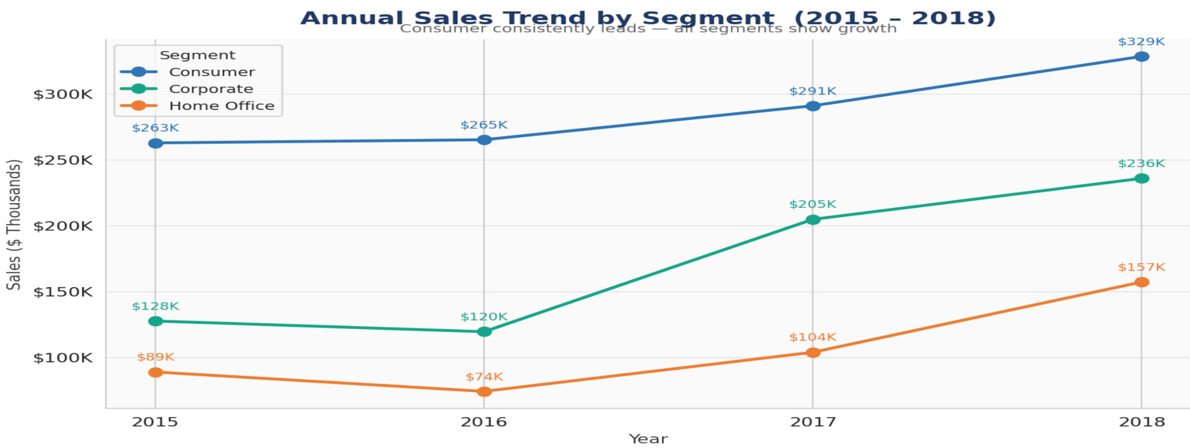
West region leads with $710K (31.4%). South region is the most underserved at 17.2% — largest untapped growth opportunity.

Chart 7 — Sales by Category & Customer Segment (Stacked)



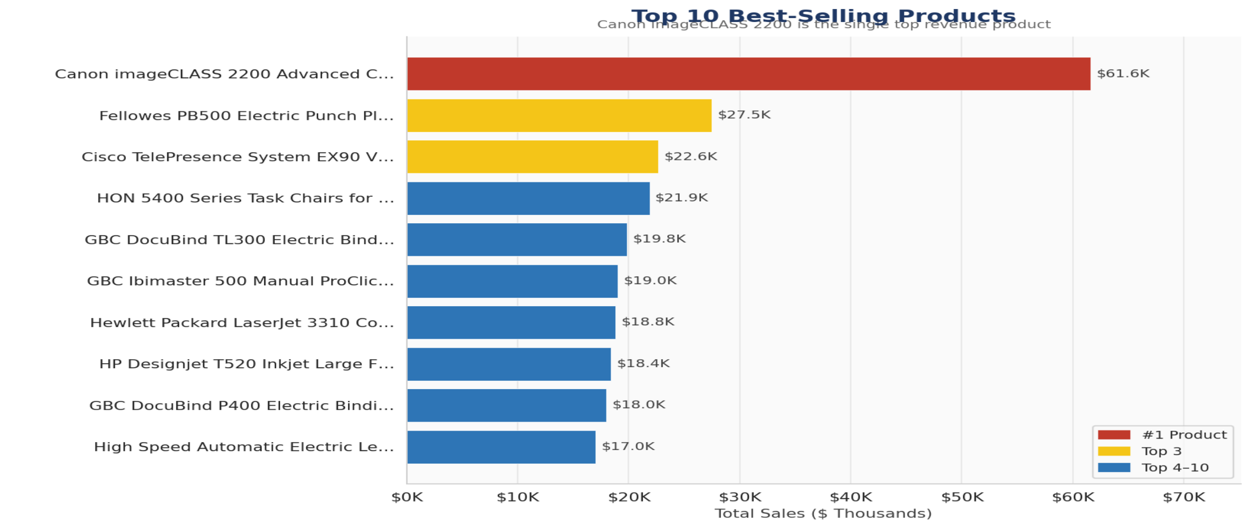
Stacked bars show segment contribution within each category. Consumer dominates across all three categories. Technology has the strongest Corporate share.

Chart 8 — Annual Sales Trend by Segment (2015–2018)



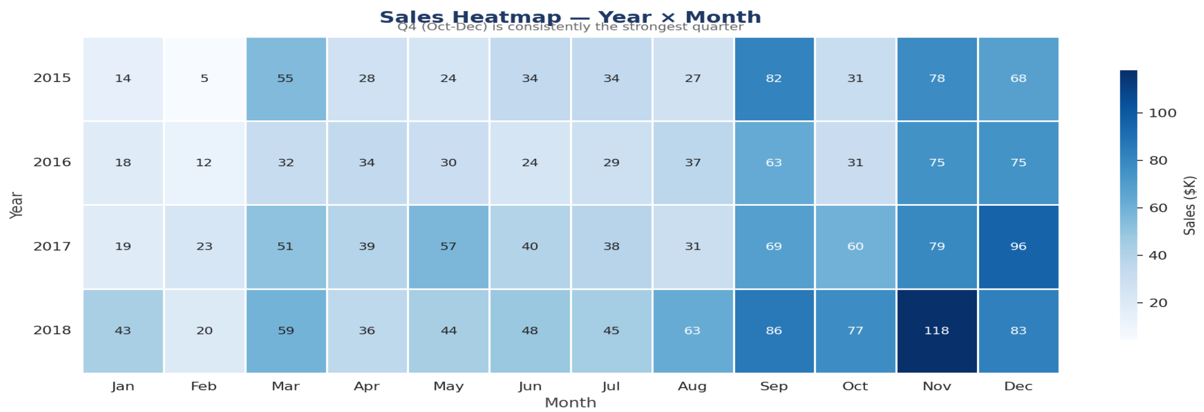
Multi-line chart showing year-over-year growth per segment. All three segments show consistent growth. Consumer leads in every year.

Chart 9 — Top 10 Best-Selling Products



Canon imageCLASS 2200 Copier is the single highest-revenue product at $61.6K. Red = #1 product, Gold = Top 3.

Chart 10 — Monthly Sales Heatmap (Year × Month)



Darker blue = higher sales. Q4 (Oct–Dec) is consistently the strongest quarter across all years. February is the weakest month in every year.

SECTION 7 — ACTIONABLE BUSINESS INSIGHTS

PATTERN 1: Consumer Segment Dominance

Consumer generates $1,148,060 — 50.8% of all revenue. This is nearly double Corporate ($688,494). Consumers place the most orders (5,101) but have the lowest Average Order Value ($225.07) vs Home Office ($243.40) and Corporate ($233.15). This classic high-volume, low-ticket pattern signals a bundling opportunity. Even a $20 increase in Consumer AOV would add $102,000 to annual revenue without acquiring a single new customer.

PATTERN 2: Technology is the Revenue Engine

Technology leads all categories at $827,455 (36.6% of total). Within it, Phones ($327,782) and Machines ($189,238) are the top 2 sub-categories overall. Copiers have extreme per-transaction value — Canon imageCLASS 2200 generated $61,599 from relatively few transactions. Technology products are ideal for Corporate and Home Office campaigns since these segments have higher AOVs and greater need for office technology solutions.

PATTERN 3: West & California Are Disproportionate Revenue Sources

West region generates $710,219 (31.4%) and California alone contributes $446,306 (19.7%) of total national revenue. California + New York together = 33.2% of revenue. Meanwhile, the South (17.2%) and Central (21.8%) are significantly underperforming relative to their population. Texas, Florida, Illinois, and Ohio show early signs of growth — these are the highest-priority expansion targets.

SECTION 8 — STRATEGIC RECOMMENDATIONS

RECOMMENDATION 1: Expand West Region Marketing & Technology Portfolio

PROBLEM: 33.2% of revenue concentrated in just 2 states (CA + NY). South and Central regions underperforming.

ACTIONS:
• Allocate 40% of marketing budget to West region — target CA Corporate and Home Office clients.
• Launch Technology Bundle campaigns (Phones + Accessories + Copiers) at 5-10% discount.
• Assign dedicated account managers to top 50 Corporate clients in California and New York.
• Run trade show campaigns in Texas and Florida to activate the South region.

EXPECTED IMPACT: 15–20% West region revenue uplift within 2 quarters. AOV projected to grow from $459 to $520+.

RECOMMENDATION 2: Rationalize Low-Performing Products & Create Bundles

PROBLEM: Fasteners ($3,001), Labels ($12,347), and Envelopes ($16,128) = only 1.4% of revenue, creating supply chain complexity with minimal return.

ACTION A — Bundle Strategy:
• Add low-value items as free gifts with orders above $500 to increase perceived value.
• Create 'Office Starter Pack' (Paper + Binders + Labels + Envelopes) at 12% discount.
• Cross-sell Storage + Tables as 'Workspace Setup' bundle.

ACTION B — Discontinuation:
• Remove Fasteners from catalog — $3,001 revenue does not justify SKU management cost.
• Redirect supplier budget to Phones and Accessories inventory.

EXPECTED IMPACT: 8–12% AOV increase. 3–5% reduction in logistics complexity costs.

SECTION 9 — EXECUTIVE SUMMARY

This comprehensive analysis of the Superstore Sales dataset (9,800 records, 2015–2018) reveals a business with strong revenue fundamentals — $2,261,536 in total sales across 4,922 orders and 793 customers. Three clear patterns emerge: Consumer segment dominance with room to raise AOV, Technology as the highest-density revenue category, and geographic concentration in California and New York. Two strategic recommendations are provided: expand West region marketing with Technology bundles, and rationalize or bundle underperforming sub-categories. Implementing both could add an estimated $180,000–$280,000 in incremental annual revenue.

Area	Finding	Recommended Action
Revenue	$2.26M across 4 years	Target 15% YoY growth
Geography	West 31.4% dominates	Expand South and Central regions
Segment	Consumer = 50.8% of sales	Upsell bundles to lift AOV
Category	Technology leads at 36.6%	Expand tech product catalog
Shipping	Standard Class = 59.3%	Market Same Day to premium clients
Products	Fasteners/Labels underperform	Bundle or discontinue low-SKUs
Seasonality	Q4 is consistently strongest	Front-load campaigns for Oct-Dec

Report prepared by: Amiin Mohamed | Lead Data Analyst | June 2026
Dataset: Superstore train.csv (Cleaned) | Tools: Python, pandas, Matplotlib, Seaborn, ReportLab